Experimental data.

Determining the laminar flame speed and emission spectra in laminar Bunsen flames

May 2026

Learning objectives:

The learning objectives of this assignment are:

- to determine the laminar flame speed of premixed methane-air and hydrogen-air flames,
- to analyse flame structure using OH-PLIF and OH* chemiluminescence imaging,
- to investigate flame emission spectra under lean and rich conditions,
- to compare the combustion characteristics of methane and hydrogen flames.

You should be able to solve the experimental part of the assignment using the lecture notes, the explanations provided during the lectures, and the explanations contained in this booklet. However, you are encouraged to consult additional literature sources and reliable online references to support your analysis and discussion. Remember to cite relevant literature if you decide to use it.

If something is unclear, please contact a.silvacorreia@tudelft.nl or f.dedomenico@tudelft.nl.

1] Introduction

Laminar flame speed is a fundamental quantity used to characterise the combustion properties of a fuel-oxidiser mixture. This quantity, S_L, is a function of pressure, temperature, equivalence ratio and reactant composition. Laminar flame speed also provides a basis for understanding turbulent flame propagation, which is important for predicting flame stabilisation and the design of combustors in gas turbine engines.

In this assignment, you will analyse the laminar flame speed of premixed methane and hydrogen flames using experimental data acquired under both lean and rich conditions.

In combustion science, optical diagnostics are powerful tools for visualising flame structures and measuring scalar and vector quantities in reacting flows. In the experiment, OH planar laser-induced fluorescence (OH-PLIF) and OH* chemiluminescence were used to visualise the flame structure and the reaction zone. In addition, a spectrometer was used to record the flame emission spectra under lean and rich conditions, to identify the radicals responsible for the different flame colours.

You have been provided with OH-PLIF and OH* chemiluminescence images to identify the flame front and analyse flame behaviour. These images will allow you to calculate the laminar flame speed of premixed hydrogen and methane flames. You will use flame emission spectra to determine the species present in the flame.

2] Experiment Description

(2.1) Laminar Flame Speed

A laminar flame typically stabilises at the location where the flow velocity balances the consumption of reactants by chemical reactions together with the diffusion of heat and chemical species across the flame front. In premixed fuel-oxidiser systems, the flow velocity component perpendicular to the flame surface balances the laminar burning velocity of the mixture. This principle is used in the Bunsen burner configuration.

A laminar premixed Bunsen flame exhibits a characteristic conical shape, as shown in Fig. 1(b). The flame cone angle adjusts such that the normal component of the burner exit velocity balances the laminar burning velocity of the mixture. As illustrated in Fig. 1(a), the cone half-angle, α, can be determined from optical images.

You can calculate the unburnt gas velocity, V_u, from the mass flow rate and density of the incoming unburnt gases and the burner cross-section. The laminar burning velocity is then obtained from:

$$S_L = V_u sin\alpha$$

In the OH-PLIF measurements acquired on the Bunsen burner, the flame front is identified by the transition between regions with negligible OH fluorescence signal and regions with strong OH fluorescence signal. This transition corresponds approximately to the reaction zone, where heat release and radical production are concentrated. In laminar premixed methane and hydrogen flames at atmospheric conditions, the flame thickness is typically of the order of a few hundred micrometres.

In the data of this assignment, the laminar Bunsen flames are imaged using OH-PLIF, a digital camera, and a UV camera equipped with a bandpass filter suitable for detecting OH* chemiluminescence. The equivalence ratio is varied while maintaining constant thermal power. You will use the acquired images to determine the laminar flame speed under these operating conditions.

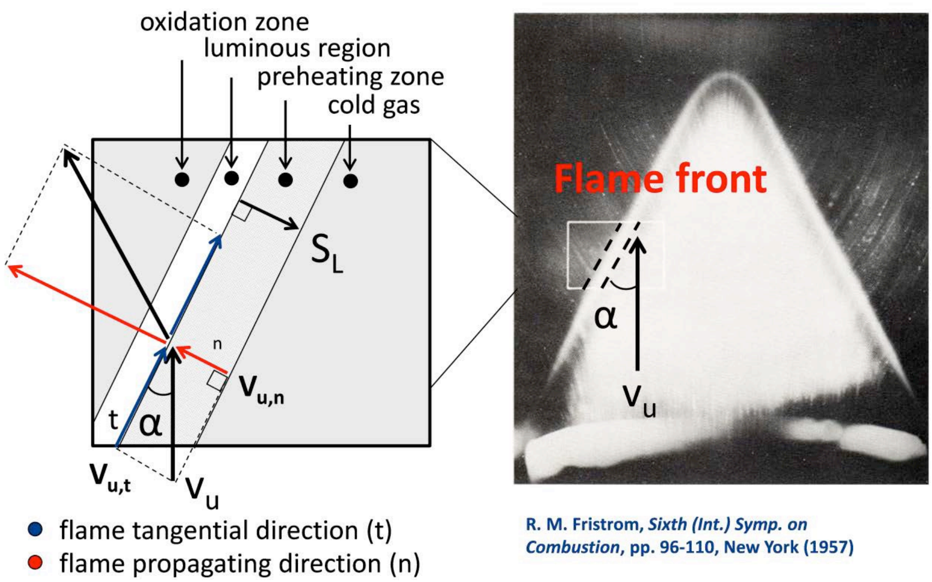

Figure 1: (left) Laminar flame speed from flow velocity components. (right) Image of a Bunsen flame

(2.2) Spectroscopic Emission Measurements

The spontaneous flame emission from methane-air Bunsen flames is measured under different operating conditions using a spectrometer. You have received the emission spectra of premixed methane-air flames.

The spectrometer collects the light emitted by electronically excited species in the flame and separates it into its constituent wavelengths. This allows the identification of specific radicals, atoms, and molecules, such as CH*, C2*, CO2*, based on their characteristic light emission features.

By analysing the measured spectra, you can gain insight into the flame composition, combustion chemistry, and radiative processes occurring during combustion. In some cases, spectral features can also provide qualitative information about flame temperature and soot formation.

3] Laboratory Setup

3.1) Bunsen burner

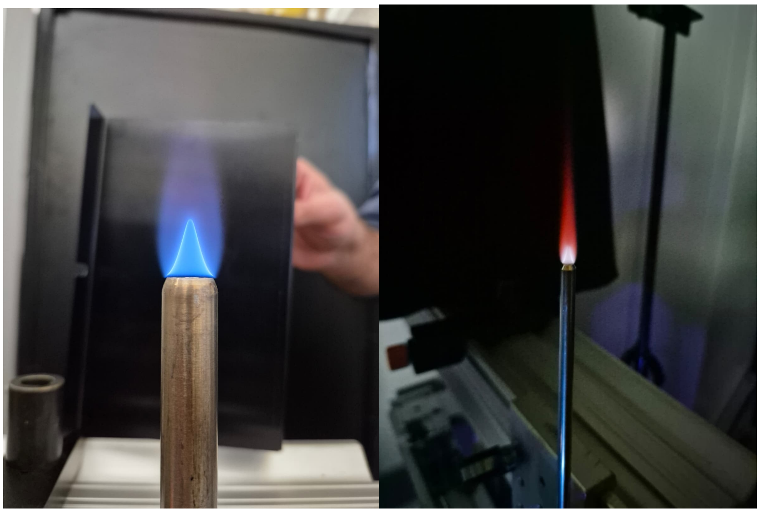

Figure 2: (left) Picture of a Methane-air premixed Bunsen burner; (right) Picture of a hydrogen-air premixed Bunsen burner

The experimental setup consists of a laminar Bunsen flame burner, as shown in Fig. 2 (methane-air flame on the left and hydrogen-air flames on the right). The reactants enter the burner through a pipe with a length of approximately 50 diameters to promote complete mixing and flow homogenisation before reaching the burner exit.

For the premixed methane flames, the Bunsen burner has an inner diameter of 21 mm. For the premixed hydrogen flames, the burner inner diameter is 4 mm.

The fuel (methane or hydrogen) is supplied from the combustion laboratory gas distribution system, which consists of pressurised gas cylinders. Compressed air is supplied by the laboratory compressor system. Both air and fuel pass through pressure regulators, control valves and mass flow controllers before entering the setup. A schematic P&ID (piping and instrumentation diagram) of the Bunsen burner is shown in figure 3.

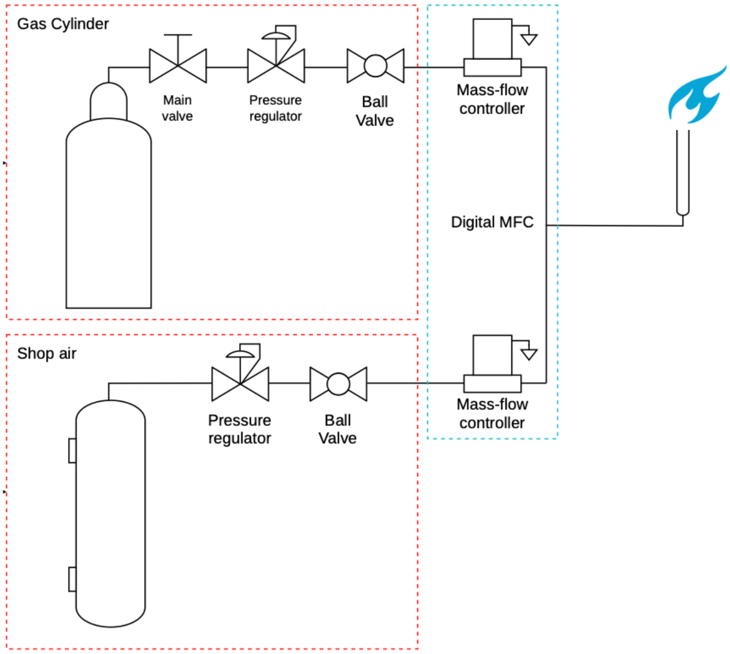

Figure 3: P&ID of the Bunsen burner experiment

Digital mass flow controllers from Bronkhorst (MFCs) are used to set the flow rates, typically expressed in NLPM (normal litres per minute) or SLPM (standard litres per minute). Although reported as volumetric flow rates, these quantities effectively correspond to mass flow rates referenced to specified temperature and pressure conditions

- **Normal conditions (ln/min):** a temperature of 0 °C and a pressure of 1013.25 hPa(a) is chosen and these reference conditions are indicated by the underlying letter 'n', in the volume unit used. The measurement method for the direct thermal mass flow is always based on these reference conditions, unless otherwise requested.

- **Standard conditions (l/min)**: in this case, the reference conditions are based on 20°C instead of 0°C. Pay attention to this difference, because mixing up these reference conditions causes a 7% error.

The mass flow controllers are designed such that they supply a constant mass flow irrespective of the incoming gas temperature and pressure.

3.2) Diagnostics

A) OH-PLIF and OH* chemiluminescence

In your lectures, you have explored the principles of PLIF and chemiluminescence measurements. In the experiment, a dye laser is used as the excitation source for the OH-PLIF measurements. Unlike a conventional fixed-wavelength laser, a dye laser is tuneable, meaning that its output wavelength can be adjusted very precisely. This is essential for laser-induced fluorescence measurements, as the laser wavelength must match a specific absorption transition of the OH radical (see Figure 4). In this experiment, the laser is tuned to a wavelength corresponding to the OH A-X (1,0) $Q_1(8)$ absorption line at 283.636 nm.

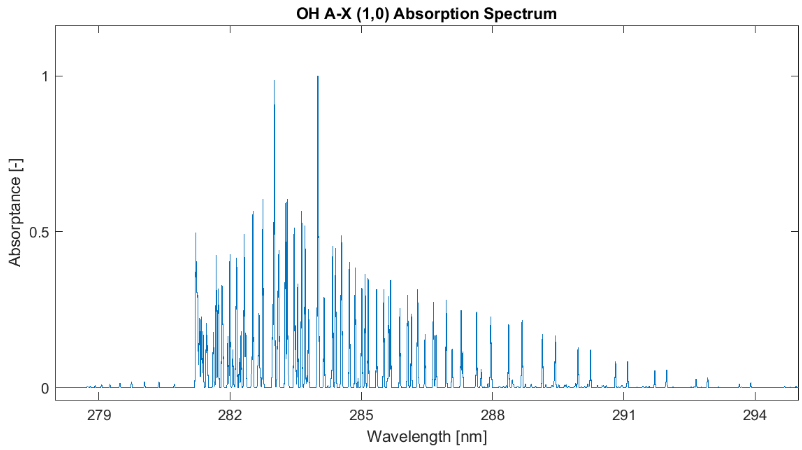

Figure 4: OH absorption spectrum within the OH A-X (1,0) transitions.

When the OH radicals absorb the laser light, they are excited to a higher electronic energy state. After a very short time, the excited molecules relax back to lower energy states and re-emit light at slightly different wavelengths. This emitted light is referred to as fluorescence, as discussed during the lectures. Since the fluorescence signal originates only from regions containing OH radicals, OH-PLIF provides images of the reaction zone of the flame.

The laser beam is shaped into a thin laser sheet using a combination of optical lenses. Typically, a collimating and cylindrical lens arrangement is used to expand the beam in one direction while maintaining a small thickness in the perpendicular direction. This creates a thin planar illumination region that intersects the flame, allowing two-dimensional imaging of the OH distribution within a selected plane of the flame.

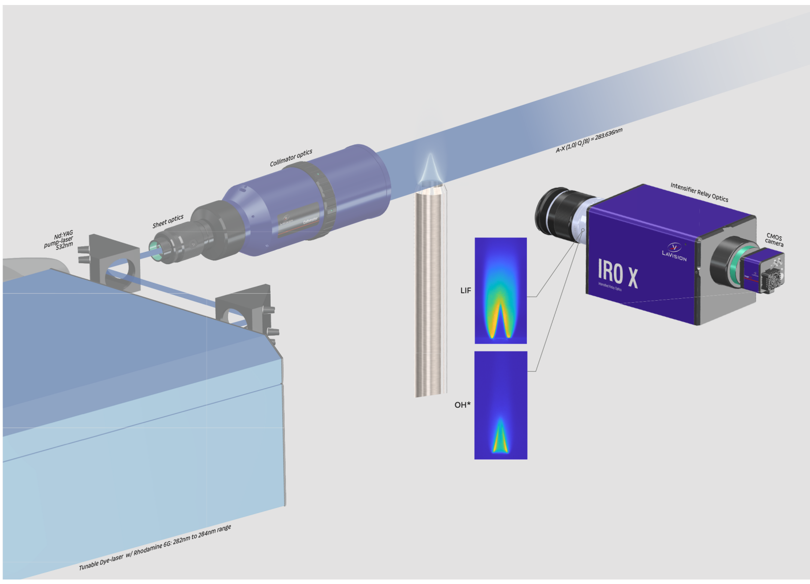

Figure 5: Schematic of the OH-PLIF measurement system

The fluorescence signal is recorded using a UV-sensitive intensified camera. A narrowband optical filter centred around the OH fluorescence wavelength (310 +- 10 nm bandwidth) is placed in front of the camera lens. The purpose of this filter is to reject scattered laser light at the excitation wavelength and isolate the weaker fluorescence emission from the OH radicals. Without this filtering, reflections and scattering of the laser beam would dominate the recorded signal. A schematic of the acquisition system used for the PLIF measurements is shown in figure 5.

The same intensified camera is also used to record OH* chemiluminescence images. In this case, the laser is turned off, and the camera exposure time is increased. OH* chemiluminescence originates from electronically excited OH radicals naturally produced during combustion reactions. The spontaneous emission around 308 nm provides a line-of-sight image of the heat-release region without requiring external laser excitation.

B) Spectrometer

A series of lenses and fiber optics, as shown in Fig. 6, is used to collect and focus the light emitted by the flame onto the entrance slit of a spectrometer (Thorlabs CCS100 Compact Spectrometer, spectral range 350–700 nm). The spectrometer contains a diffraction grating that spatially separates the incoming light into its constituent wavelengths. The dispersed light is then recorded by a CCD detector, enabling measurement of the flame emission spectrum with sufficient spectral resolution to distinguish important combustion radicals such as CH* and C2*.

The optical collection system increases the amount of light coupled into the spectrometer and improves the signal-to-noise ratio of the measurements. Each chemical species emits radiation at characteristic wavelengths corresponding to specific electronic or vibrational transitions. By analysing the measured spectrum, it is therefore possible to identify the excited species present in the flame.

Figure 6: Schematic picture of the optical arrangement used for acquiring the flame emission spectra.

(3.3) Mass flow calculations

Suppose that an air mass flow rate of $1\, g/s$ is required. The mass flow controllers used in this experiment display the flow rate in normal litres per minute (NLPM), corresponding to the volumetric flow rate referenced to normal conditions (NTP).

For the air mass flow controllers used in this setup, the normal conditions are defined as:

- Temperature: 0° ($273.15\, K$)
- Pressure: $1\, atm$

Under these conditions, the density of air is approximately ρ_{air} = 1.293 kg/m^3.

The equivalent volumetric flow rate at normal conditions can therefore be calculated from$V_n = \dot{m}/\rho_n$, where: m˙ is the mass flow rate, and ρ_n is the density at normal conditions.

Therefore, an air mass flow rate of $1\, g/s$ corresponds to approximately to 1e-3/1.293*60000 nlpm = $46.4\, NLPM$ under the specified normal conditions.

<u>The temperature and pressure in the laboratory during the experiment could be taken as 300 K and 1 atm respectively.</u>

4] Experimental procedure followed to acquire the experimental data

(4.1) Scaling and Calibration

1) The camera is focused to the central plane of the burner pipe.
2) Pictures of the setup with some mm paper for the purpose of scaling are taken.

(4.2) Start up procedure

1) The required fuel and air value on the air mass flow controller are set.
2) When the fuel flow is stable, the flame is ignited with a spark.
3) After successful ignition, some time is left for the flow to stabilize before acquiring the data.

(4.3) Burner operation

1) All the experiments are done at constant thermal power.
2) The OH* images and the spectrometer emission data of the flame are acquired.
3) Laser is turned on to acquire the OH-PLIF images.
4) When flashback and blowoff limits occurr, the fuel is shut down immediately.

5] Preparation for the assignment: Generation of the test matrixes

Premixed flames: You will explore the flame behavior for premixed methane-air flames and premixed hydrogen-air flames

a. You should calculate the air and fuel mass flow required for the methane premixed flame at constant thermal power of **1.00 kW** for equivalence ratios (Φ) from 0.8 to 1.2 by increasing the equivalence ratio by 0.2 each step.

b. You should calculate the air and fuel mass flow required for the hydrogen premixed flame at constant thermal power of **0.250 kW** for equivalence ratios (Φ) from 0.5 to 1.3 by increasing the equivalence ratio by 0.1 each step.

H2 Premixed flame setpoints	
Φ	Stability
< 0.5	Blow-off
0.5	
0.6	
0.7	
0.8	
0.9	
1.0	
1.1	
1.2	
1.3	
>1.55	Flashback

CH4 Premixed flame setpoints	
Φ	Stability
< 0.8	Blow-off
0.8	
1.0	
1.2	
> 1.2	Flashback

Table 1: (a) H2 and (b) CH4 setpoints and stability at P = 250W and 1kW respectively.

You should make separate tables in Excel for the experimental condition variations mentioned in 5], reporting the following quantities for the experiment conditions:

1) Fuel flow rate (kg/s)
2) Fuel flow rate (nlpm)
3) Air flow rate (kg/s)
4) Air flow rate (nlpm)
5) Thermal Power (kW)
6) Pipe flow density (kg/m3)
7) Pipe Reynolds number (-)
8) Pipe bulk velocity (m/s)

6] Data for the assignment

The provided data is organized into a main folder containing four subfolders, along with MATLAB and *Python* scripts for reading *.b16 files. The first subfolder, *LIF/*, contains directories for each experimental conditions following *[Fuel]_PHI[EquivalenceRatio]/* format. Each of these directories includes a *B16/* subfolder which stores the corresponding *.b16 files. The second subfolder, *OHstar/* follows the same structure. The third subfolder, Calibration/, contains the calibration images acquired for each fuel *[Fuel]/B16/*.b16*. These calibration images were acquired using metric paper, where each small square corresponds to 1mm. The fourth subdfolder contains the images acquired from the spectrometer for premixed methane flames *[Fuel]_PHI[EquivalenceRatio].*

- **Flame imaging**: you have received the average of 200 snapshots of OH PLIF and OH* chemiluminescence of the flames using the UV intensified camera (examples in figure 7 and 8).

- **Spectral measurements**: you have received snapshots of the emission spectra of the premixed methane flames for phi = 0.8, 1, and 1.2.

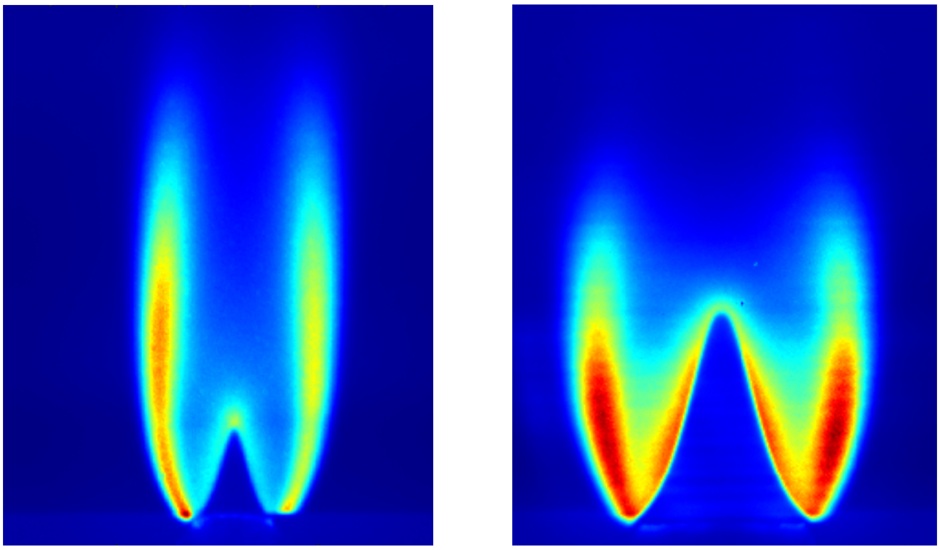

Figure 7: Acquired OH PLIF images for (left) H2 and (right) CH4 at φ = 1 at constant thermal power of 250 W and 1 kW respectively.

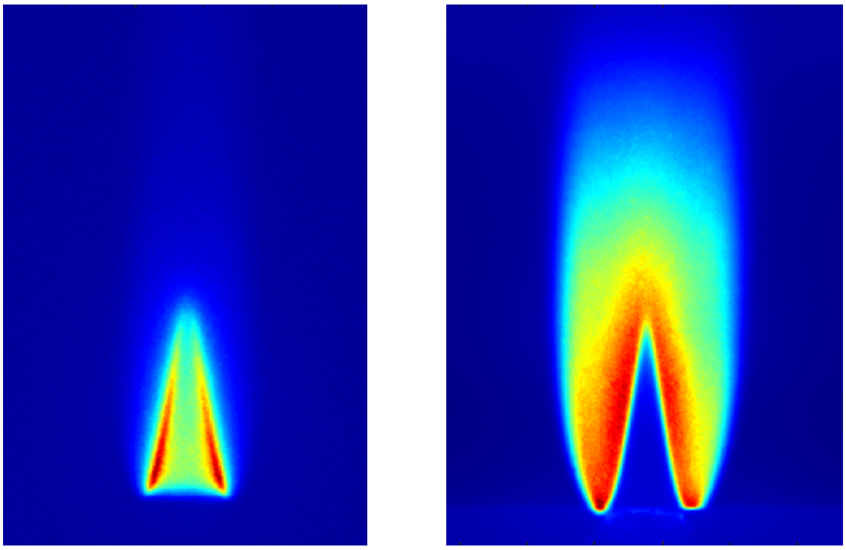

Figure 8: Acquired (left) OH and (right) OH PLIF images for H2 at φ = 0.7 at constant thermal power of 250W respectively.*

Abel inversion:

OH* is an excited combustion radical formed during chemical reactions involving the OH molecule. When OH* relaxes back to its ground state, it emits a photon, a process known as chemiluminescence. Because this emission is spontaneous and occurs throughout the flame volume, any OH* image represents a line-of-sight integration of the entire 3D flame onto a 2D plane. As a result, the recorded intensity at each pixel includes contributions from all emitting regions along the viewing direction.

This makes direct interpretation of OH* chemiluminescence difficult. For example, when plotting the intensity profile of OH* across the flame, the reactant side does not appear to go to zero because the emission from other regions along the line of sight is still captured.

However, if the flame is axisymmetric, this limitation can be overcome by mathematically reconstructing the local (radial) emission distribution from the integrated signal. This reconstruction is performed using Abel inversion, which transforms the measured line-of-sight integrated intensity into a radial emissivity distribution representing a cross-section of the flame. In other words, Abel inversion allows reconstruction of the internal radial structure of an axisymmetric flame from its projected image. Thus, Abel inversion allows reconstruction of the internal radial structure of an axisymmetric flame from its projected image.

Abel inversion relates the measured intensity $I(x)$ to the local emissivity $\varepsilon(r)$ through:

$$I(x) = 2\int_{x}^{\infty} \frac{\varepsilon(r)r}{\sqrt{r^2 - x^2}} dr$$

The inversion procedure then recovers the local emissivity distribution $\varepsilon(r)$ from the measured intensity distribution $I(x)$.

AE4262 Laboratory Exercise 2026 -- Annotated Technical Guide

This document accompanies the original Laboratory Exercise PDF. Annotations are derived from lecture transcripts (20 May and 21 May 2026) and cross-referenced with the Warnatz combustion textbook, the Ulrich Maas lecture notes, and the Lecture Notes L1-L2.

Section 1 -- Introduction (Laboratory Exercise pp. 1)

The laminar flame speed S_L is a physicochemical property of a specific fuel-oxidizer mixture at given pressure and temperature. It is defined as the velocity at which a planar, adiabatic, unstretched flame front propagates into the unburnt mixture in the direction normal to itself.

[Lecture 5 slides, p. 23] [Maas notes, p. 5]

S_L depends on pressure, temperature, equivalence ratio, and fuel composition. In gas turbine engine design, S_L is a fundamental parameter for predicting flame stabilization, flashback risk, and turbulent flame propagation models (e.g., the Damkohler model vT = vL + v', where v' is the turbulent velocity fluctuation).

[Maas notes, p. 48, Damkohler model] [Lecture 9 slides, turbulent burning velocity]

Section 2 -- Experiment Description Annotations

2.1 Laminar Flame Speed Measurement

The Bunsen burner cone-angle method relies on the kinematic balance at a stationary flame front. At the flame surface, the incoming flow velocity V_u has a component normal to the flame (V_u sin alpha) and a component tangential to the flame (V_u cos alpha). For a steady flame, the normal component equals S_L

$$S_L = V_u \sin(\text{alpha})$$

[Laboratory Exercise, Section 2.1] [Maas notes, p. 5, Bunsen flame relation] [Warnatz, Chapter on premixed flames]

The unburnt gas velocity V_u is calculated from the total mass flow rate, the mixture density, and the burner cross-section area A

$$V_u = \text{m\_dot\_total} / (\text{rho\_mix} * A)$$

The OH-PLIF imaging technique identifies the flame front as the sharp gradient in OH fluorescence intensity. OH is an intermediate radical that is virtually absent in the cold unburnt gas and reaches high concentration in the reaction zone and post-flame region. The spatial resolution of OH-PLIF is sufficient to resolve the flame front position to within the laser sheet thickness (approximately 0.5 mm).

[Lecture 7 slides, PLIF section] [Lecture 6 slides, LIF diagnostics]

2.2 Spectroscopic Emission Measurements

The spectrometer collects spontaneous emission from electronically excited radicals. The key species detectable in the 350-700 nm range for methane-air flames are CH* (431 nm), C_2* (Swan bands at 473, 516, 556 nm), and the broadband CO_2* continuum. OH* emission near 308 nm falls below the spectrometer lower bound.

[Lecture 6 slides, chemiluminescence section] [Laboratory Exercise, Section 2.2]

> During the 20 May lecture, the instructor demonstrated the spectrometer on a live methane-air Bunsen flame. As the equivalence ratio was increased from lean toward rich, the green C_2* bands appeared and intensified. When the flame became sufficiently rich, broadband soot luminosity (black-body radiation from hot particles) dominated the spectrum. The instructor noted that soot radiation is fundamentally different from chemiluminescence -- soot particles emit as approximate black bodies with a continuous spectrum peaked around 600-700 nm, while chemiluminescence arises from discrete electronic transitions in gas-phase radicals.

[Lecture transcript 20 May, spectrometer demonstration]

Section 3 -- Laboratory Setup Annotations

3.1 Bunsen Burner Configuration

The methane burner inner diameter is 21 mm. The hydrogen burner inner diameter is 4 mm. The fuel and air are premixed upstream of the burner exit through a pipe of approximately 50 diameters length. This long entry length ensures a fully developed velocity profile (parabolic Poiseuille profile for laminar flow) and complete homogenization of the fuel-air mixture before it reaches the burner exit plane.

[Laboratory Exercise, Section 3.1]

The 50-diameter mixing length is standard practice for ensuring that the premixing is complete. For laminar pipe flow, the hydrodynamic entry length is approximately L_entry = 0.06 Re D. For Re = 1000 and D = 21 mm, L_entry approximately equals 1.26 m, or about 60 diameters. The actual pipe length is therefore close to ensuring full flow development at moderate Reynolds numbers.

3.2 OH-PLIF System Details

The dye laser is tuned to 283.636 nm, targeting the OH A-X (1,0) Q_1(8) absorption line. This transition was chosen because it has a relatively temperature-insensitive absorption cross-section, reducing the sensitivity of the fluorescence signal to local temperature variations across the flame.

[Laboratory Exercise, Section 3.2A] [Lecture 7 slides, LIF technique]

The fluorescence is collected through a bandpass filter centered at 310 +/- 10 nm. This filter rejects the scattered excitation laser light at 283.636 nm and passes only the red-shifted fluorescence from the OH (1,1) and (0,0) emission bands near 310 nm. Without this filter, laser scattering from optical surfaces and burner components would overwhelm the much weaker fluorescence signal.

[Laboratory Exercise, Section 3.2A] [Lecture transcript 20 May, PLIF system explanation]

The same intensified camera acquires OH* chemiluminescence images with the laser off and a longer exposure time. The OH* spontaneous emission at approximately 308 nm is captured through the same bandpass filter. Because no laser excitation is used, the signal is volumetric (line-of-sight integrated) rather than planar.

3.3 Mass Flow Controller Details

The Bronkhorst digital MFCs report flow in NLPM (normal litres per minute) referenced to normal conditions (0 deg C, 1013.25 hPa). The conversion formula is

```
Q_n = m_dot / rho_n * 60000 [NLPM]
```

where rho_n is the density of the specific gas at normal conditions. For air, rho_n = 1.293 kg/m^3. For CH_4, rho_n = 0.7138 kg/m^3. For H_2, rho_n = 0.08921 kg/m^3.

[Laboratory Exercise, Section 3.3]

> Standard conditions (SLPM) use 20 deg C as the reference temperature instead of 0 deg C. This causes approximately 7% discrepancy in the reported volumetric flow for the same mass flow rate (because rho at 20 deg C is lower by a factor of 273.15/293.15 = 0.932). The MFCs in this experiment use normal conditions.

Section 4 -- Experimental Procedure Annotations

The constant thermal power constraint requires careful treatment of the limiting reactant. The thermal power is P_th = m_dot_fuel_reacted * LHV, where m_dot_fuel_reacted is the mass flow of fuel that actually participates in combustion. For lean and stoichiometric conditions (phi <= 1), all fuel burns and the fuel is the limiting reactant. Therefore m_dot_fuel = P_th / LHV is fixed, and m_dot_air = m_dot_fuel * s_air / phi decreases as phi increases toward stoichiometry.

For rich conditions (phi > 1), the oxidiser (air) becomes the limiting reactant. Only the fraction of fuel that the available oxygen can combust contributes to heat release. To maintain the same thermal power, the air flow must equal the stoichiometric air flow (since all the oxygen must be consumed to release P_th of energy). Thus m_dot_air is held constant at its stoichiometric value, m_dot_air = P_th * s_air / LHV. The fuel flow then increases with phi as m_dot_fuel = m_dot_air * phi / s_air. The excess fuel passes through unburned.

[Laboratory Exercise, Section 4.3] [Lecture transcript 20 May]

Physically, for lean mixtures the bulk velocity decreases as phi increases because the air flow decreases while fuel flow stays constant. At stoichiometry the bulk velocity reaches a minimum for the lean branch. Beyond phi = 1 the air flow is locked and extra fuel is added, so the bulk velocity stays approximately constant (increasing only slightly from the added fuel mass). Combined with the fact that laminar flame speed peaks near stoichiometry and drops for very rich mixtures, this means flashback occurs not because the velocity drops further but because the flame speed may still exceed the (now roughly constant) bulk velocity. The observed flashback limit for hydrogen is above phi = 1.55, and for methane above phi = 1.2.

[Laboratory Exercise, Table 1]

Section 5 -- Test Matrix Generation Annotations

5.1 Methane Test Matrix

For methane at P_{th} = 1.00 kW with LHV = 50.0 MJ/kg

At stoichiometry and for lean conditions, the fuel mass flow is fixed by m_dot_fuel = P_th / LHV = 1000 / (50.0 x 10^6) = 2.000 x 10^{-5} kg/s. The stoichiometric air-to-fuel mass ratio for CH_4 is s_air = (2 * 32 * 4.76) / (1 * 16) = 17.167.

For phi <= 1 (lean), m_dot_air = m_dot_fuel * s_air / phi. For phi > 1 (rich), m_dot_air is held constant at the stoichiometric value m_dot_air,st = 2.0 x 10^{-5} * 17.167 = 3.433 x 10^{-4} kg/s, and the fuel flow increases as m_dot_fuel = m_dot_air,st * phi / s_air = 2.0 x 10^{-5} * phi.

phi	m_dot_fuel [kg/s]	m_dot_air [kg/s]	Q_fuel [NLPM]	Q_air [NLPM]	V_bulk [m/s]
0.8	2.00e-5	4.29e-4	1.68	19.92	1.12
1.0	2.00e-5	3.43e-4	1.68	15.93	0.90
1.2	2.40e-5	3.43e-4	2.02	15.93	0.91

Notice that at phi = 1.2 the air flow equals the stoichiometric value (same as phi = 1.0) while the fuel flow has increased by a factor of 1.2. The bulk velocity at phi = 1.2 is therefore nearly identical to that at phi = 1.0, unlike the lean side where reducing phi increases air flow and hence velocity. Approximate values shown. The Reynolds number for each condition must be computed as Re = rho_mix * V_bulk * D / mu_mix. At laboratory conditions (300 K, 1 atm), the mixture density is approximately rho_mix approximately equal to 1.15-1.18 kg/m^3 (close to air density since the fuel mass fraction is small).

5.2 Hydrogen Test Matrix

For hydrogen at P_{th} = 0.250 kW with LHV = 120.0 MJ/kg

m_dot_H2,st = 250 / (120.0 x 10^6) = 2.083 x 10^{-6} kg/s. s_air = (0.5 * 32 * 4.76) / (1 * 2) = 34.333. The stoichiometric air flow is m_dot_air,st = 2.083 x 10^{-6} * 34.333 = 7.153 x 10^{-5} kg/s.

For phi <= 1 (lean), m_dot_fuel = 2.083 x 10^{-6} kg/s (constant) and m_dot_air = m_dot_fuel * s_air / phi. For phi > 1 (rich), m_dot_air = 7.153 x 10^{-5} kg/s (constant, equal to stoichiometric air) and m_dot_fuel = 2.083 x 10^{-6} * phi.

phi	m_dot_fuel [kg/s]	m_dot_air [kg/s]	Q_fuel [NLPM]	Q_air [NLPM]	V_bulk [m/s]
0.5	2.08e-6	1.43e-4	1.40	6.64	9.96
0.6	2.08e-6	1.19e-4	1.40	5.53	8.32
0.7	2.08e-6	1.02e-4	1.40	4.74	7.15
0.8	2.08e-6	8.94e-5	1.40	4.15	6.28
0.9	2.08e-6	7.95e-5	1.40	3.69	5.59
1.0	2.08e-6	7.15e-5	1.40	3.32	5.05
1.1	2.29e-6	7.15e-5	1.54	3.32	5.06
1.2	2.50e-6	7.15e-5	1.68	3.32	5.08
1.3	2.71e-6	7.15e-5	1.82	3.32	5.09

Approximate values. For phi > 1 the air flow is locked at the stoichiometric value and the fuel flow increases with phi. This causes the bulk velocity to plateau rather than continue decreasing. The bulk velocities for hydrogen are significantly higher than for methane despite lower total mass flow because the burner cross-section is much smaller (A_H2 = 1.257 x 10^{-5} m^2 versus A_CH4 = 3.464 x 10^{-4} m^2, a factor of approximately 27.5). This design choice ensures V_bulk > S_L for hydrogen at all stable conditions.

[Laboratory Exercise, Section 5 and Table 1]

Section 6 -- Data Processing Annotations

6.1 Image Calibration

The calibration images acquired with millimeter paper allow conversion from pixel coordinates to physical coordinates (mm). The calibration procedure involves identifying the spacing of the millimeter grid lines in the image and computing a scale factor (mm per pixel). This scale factor is applied to all subsequent flame images to extract physical dimensions such as the cone height and half-angle.

[Laboratory Exercise, Section 4.1]

6.2 Reading B16 Files

The .b16 files are a binary image format used by the intensified camera system. MATLAB and Python scripts for reading these files are provided with the data package. Each .b16 file contains a single 2D image. The provided data represents the average of 200 individual snapshots, which reduces shot noise and provides a representative mean flame structure.

[Laboratory Exercise, Section 6]

6.3 Abel Inversion Implementation

The Abel inversion transforms line-of-sight integrated OH* chemiluminescence images into radial emissivity profiles for axisymmetric flames. The mathematical relation is

```
epsilon(r) = -(1/pi) * integral_r^R (dI/dy) / sqrt(y^2 - r^2) dy
```

[Laboratory Exercise, Abel inversion section]

In practice, numerical implementations are used. The instructor noted that Python packages for Abel inversion exist (e.g., PyAbel) and can be used directly. Alternatively, the three-point or onion-peeling method provides a discrete implementation. The key requirement is that the flame must be axisymmetric. The flame axis (centerline) must be identified in the image, and only one half of the symmetric image is needed (the other half can be averaged with it to reduce noise).

[Lecture transcript 20 May, Abel transform discussion]

After Abel inversion, the reconstructed OH* emissivity profile should be compared with the OH-PLIF image at the same equivalence ratio. The expected differences are as follows. The Abel-inverted OH* profile peaks sharply at the flame front (the reaction zone) because OH* chemiluminescence is proportional to the local heat release rate. The OH-PLIF signal, measuring ground-state OH concentration, shows a broader profile that extends well downstream of the flame front into the post-flame equilibrium zone. A 1D flame simulation can confirm this by plotting the heat release rate and OH mole fraction as functions of the axial coordinate -- the heat release rate peak is narrow and coincident with the reaction zone, while the OH concentration rises sharply at the front but then plateaus or slowly decays in the burned gas.

[Lecture transcript 20 May, PLIF vs chemiluminescence comparison] [Lecture 6 slides, chemiluminescence section] [Maas notes, p. 29, flame structure profiles]

Reference Summary

Property	CH₄	H₂	Source
LHV [MJ/kg]	50.0	120.0	Warnatz, Ch. 1
W [g/mol]	16.04	2.016	Standard
nu_O2 (stoich.)	2	0.5	Balanced reaction
s_air	17.17	34.33	Lecture Notes L1-L2
S_L,stoich [cm/s]	~37	~200	Maas notes, p. 29
delta_q [mm]	2.5	0.64	Lecture 5 slides, p. 38
D_burner [mm]	21	4	Lab Exercise, Sec. 3.1
P_th [kW]	1.00	0.250	Lab Exercise, Sec. 5
T_ad,stoich [K]	~2222	~2380	Maas notes, p. 18
rho_n [kg/m3]	0.714	0.089	Ideal gas at NTP

All numerical values are approximate and should be verified against the specific mechanism file and thermodynamic database used in the Cantera or CEA calculations. The JANNAF polynomial coefficients are the authoritative source for thermodynamic properties.